

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **predictive machine learning models** using **Python (scikit-learn, pandas, NumPy)** for healthcare payment anomaly detection, achieving **94% precision** and reducing false-positive fraud alerts by **31%** across 2.5M+ monthly transactions in production.
- Built end-to-end **data pipelines and ETL workflows** using **Python, SQL (PostgreSQL, Oracle, MySQL)**, and **Apache Spark** to ingest, transform, and aggregate multi-source healthcare claims data; optimized pipeline performance by **45%** through vectorized **pandas** operations and partitioned **Spark RDDs**.
- Developed and published **Power BI dashboards and interactive Tableau visualizations** for healthcare operations and finance stakeholders, delivering real-time insights into claims processing KPIs, provider network performance, and cost-trend analysis; dashboards automated reporting for **15+ executives**, eliminating **8 hours/week** of manual reporting.
- Conducted statistical analysis and **A/B testing** on healthcare payment workflows using **R, hypothesis testing, and regression modeling**; validated claims-routing algorithm changes with **95% confidence intervals**, resulting in **\$2.3M annual cost savings** and **12% improvement** in claims settlement time.
- Managed **data quality and validation workflows** across healthcare systems; authored **SQL queries** for audit trails and reconciliation logic, ensuring **HIPAA, GDPR, and regulatory compliance** in production claims and payment databases; documented data-governance standards in **Confluence** for cross-functional teams.
- Automated **data science workflows and model deployment** using **Python, GitHub Actions, and AWS Lambda**; orchestrated **CI/CD pipelines** for retraining and serving predictive models at scale, reducing time-to-insight by **50%** and enabling real-time decision-support in claims adjudication systems.

SKILLS

Data Science & Machine Learning – predictive modeling, supervised learning, unsupervised learning, classification, regression, anomaly detection, model evaluation and tuning, scikit-learn, TensorFlow.

Data Engineering & Pipeline Development – ETL workflow design, Apache Spark, data warehousing, data transformation, data validation, schema design, PostgreSQL, Oracle, MySQL.

Programming & Scripting – Python, R, SQL, shell scripting, Bash, Java, REST API integration.

Business Intelligence & Analytics – Power BI, Tableau, DAX, Power Query, data modeling, dashboard development, statistical analysis, A/B testing, hypothesis testing, time-series analysis.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, GitHub Actions, CI/CD pipelines.

Healthcare & Regulatory Domain – HIPAA compliance, GDPR, healthcare claims processing, payment workflows, provider network analytics, cost analytics, claims adjudication systems.

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence, Linux, Vercel, Apify.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

PROJECTS

Healthcare Claims Anomaly Detection & Predictive Analytics (HCL Technologies)

- Built and deployed **machine learning classification model** using **scikit-learn (Random Forest, Gradient Boosting)** to detect fraudulent and erroneous claims in real-time; trained on **500K+ labeled claims records** with **feature engineering in pandas**, achieving **94% precision, 89% recall**, and preventing **\$8.2M in invalid payouts annually**.
- Designed **Apache Spark-based ETL pipeline** to aggregate claims data from disparate sources (ERP systems, claims databases, provider networks); transformed **15+ TB monthly claims volume** using **PySpark DataFrames**, applied schema validation and **data quality checks**, and loaded into analytical warehouse in **PostgreSQL**.
- Created **Power BI dashboards** for healthcare finance and operations teams displaying claims-processing metrics, fraud trends, and provider performance; embedded **DAX measures** for cost-per-claim analysis and **time-series forecasting**; dashboards served **25+ daily active users** and reduced manual reporting overhead by **12 hours/week**.
- Conducted **statistical hypothesis testing** and **A/B testing** using **R (ggplot2, dplyr)** to validate claims-routing algorithm changes; delivered results with **95% confidence intervals**, supporting **\$2.3M cost optimization** initiative across 8-state provider network.

Ireland Jobs Tracker – AI-Powered Resume Generation & Workflow Automation

- Engineered end-to-end **AI workflow orchestration system** integrating **Claude API (Anthropic) via Python SDK** to generate JD-tailored resumes at scale; implemented **prompt engineering** with structured JSON constraints and **guardrails**, processing **200+ job descriptions** and generating **personalized resume variants** in batch mode.
- Architected **data pipeline automation** using **Apify scrapers** for job-description ingestion, **Python data parsing**, Claude API integration, and **reportlab** for PDF generation; orchestrated workflow via **GitHub Actions CI/CD** with automated job triggers and **Vercel serverless deployment**.
- Demonstrated mastery of **LLM prompt engineering techniques** including few-shot prompting, constraint specification, and output validation to ensure consistency and relevance; reduced resume generation latency by **40%** through asynchronous API calls and batch processing, enabling real-time AI-powered job-matching automation.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)